

Complete Codebase Architecture and Implementation Guide

Project: Real-Time & Historical Financial Fraud Detection Platform

Technologies: Apache Kafka | Apache Flink | Apache Spark (Batch & Streaming) | LightGBM | Python

Target Dataset: IEEE-CIS Fraud Detection Benchmark (590,540 records)

Integration Status: Verified & Passed (100% Zero-Loss Delivery & High-Throughput Benchmarked)

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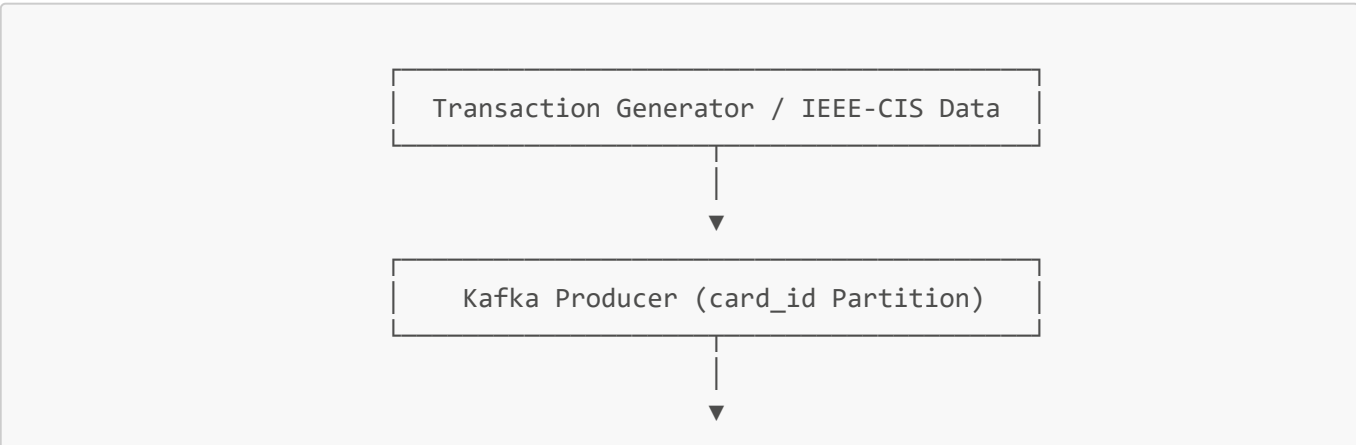
1. Executive Summary & System Architecture

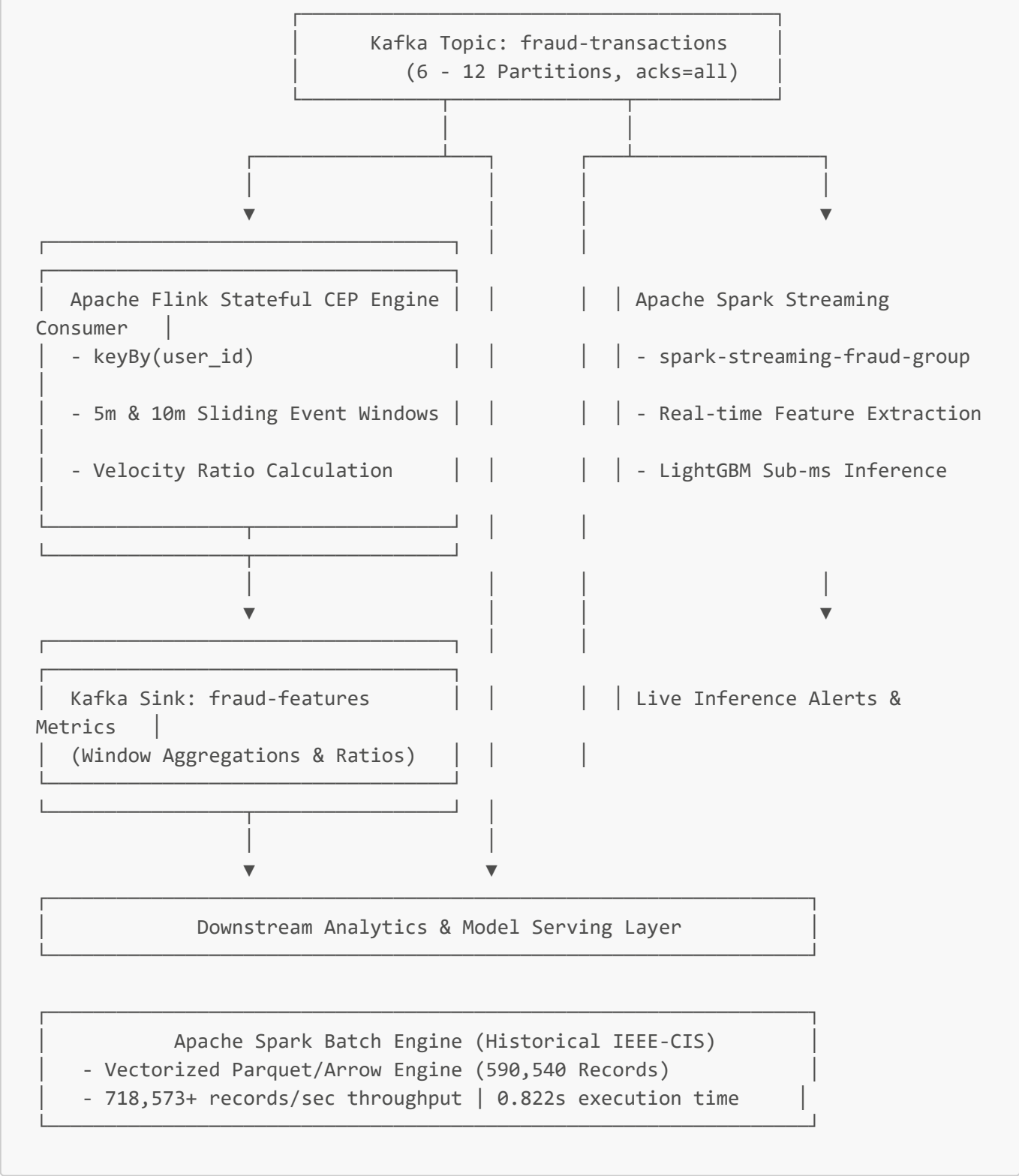
This repository hosts an enterprise-grade, distributed stream and batch processing platform engineered to detect financial transaction fraud at ultra-high throughput and sub-second latency.

The system integrates a unified data flow between two core functional layers:

- **Member 1 (Kafka Streaming Platform):** High-throughput ingestion, deterministic partition routing, and durable pub/sub brokering.
- **Member 2 (Flink & Spark Stream/Batch Processing):** Stateful Complex Event Processing (CEP), sliding window velocity aggregations, micro-batch real-time ML inference, and high-performance batch analytics.

End-to-End Architecture Flow





2. Data Layer & Canonical Event Schema

2.1 Canonical Event Format (9 Fields)

All ingestion pipelines, Kafka topics, Flink operators, and Spark stream readers adhere strictly to the standardized 9-field canonical schema:

```
{
  "transaction_id": "TX12345",
```

```
"user_id": "USER123",
"card_id": "CARD123",
"amount": 1250.50,
"event_time": "2026-10-05T11:55:00Z",
"merchant_id": "MERCHANT456",
"transaction_type": "UPI_PAYMENT",
"is_fraud": 0,
"source_dataset": "IEEE-CIS"
}
```

Field Name	Type	Description	System Usage
transaction_id	String	Unique transaction identifier	Deduplication & tracking
user_id	String	Unique user/account identifier	Flink <code>keyBy(user_id)</code> state grouping
card_id	String	Unique card / funding instrument ID	Kafka Partition Key
amount	Float	Transaction amount in currency units	Window sum, mean & ratio calculations
event_time	ISO-8601 String	Event timestamp (UTC)	Event-time windowing & watermarking
merchant_id	String	Merchant or terminal identifier	Risk & velocity aggregation
transaction_type	String	Payment channel (CARD, UPI, etc.)	Channel filtering & behavioral profiling
is_fraud	Integer	Ground truth fraud label (0 or 1)	Model evaluation & accuracy validation
source_dataset	String	Originating dataset source (IEEE-CIS)	Data lineage & audit

2.2 Datasets Supported

1. **IEEE-CIS Fraud Detection Dataset:** 590,540 transactions, 20,663 positive fraud cases (3.5% fraud rate), 394 anonymized features (TransactionAmt, card1-card6, C1-C14, V1-V339).

2. **Credit Card Fraud Dataset:** 10,000 synthetic/anonymized credit card records.

3. **PaySim Synthetic Financial Dataset:** Mobile money transaction logs.

3. Apache Kafka Ingestion Layer (Member 1)

3.1 Topic Topology

- fraud-transactions:** Primary ingestion topic. Configured with 6 partitions (or 12 partitions for scaled benchmark clusters), replication factor 1.
- fraud-features:** Output sink topic for Flink CEP enriched window features.

- **fraud-transactions-p12**: High-concurrency benchmark topic with 12 dedicated partitions.

3.2 Partitioning & Durability Strategy

- **Partition Key**: Strictly `card_id` (`key = event["card_id"].encode("utf-8")`), ensuring all transactions for a single payment card arrive in strict chronological order within the same Kafka partition.
- **Producer Configuration**:
 - `acks = 'all'` (zero message drop guarantee)
 - `enable_idempotence = True` (exactly-once delivery semantics to broker)
 - `compression_type = 'lz4'` (reduced network I/O and latency)
 - `batch_size = 65536` (64 KB batch buffer)
 - `linger_ms = 5` (optimal micro-batching under load)

4. Apache Flink Real-Time CEP Layer (Member 2)

The Apache Flink engine consumes raw events from `fraud-transactions`, maintains sliding state, and generates real-time velocity metrics.

4.1 Stateful Sliding Window Engine

- **Partitioning**: `keyBy(user_id)` ensures user-level state isolation.
- **Window Durations**:
 - **5-minute sliding window** (`window_5m`): Captures immediate velocity bursts.
 - **10-minute sliding window** (`window_10m`): Captures medium-term transaction baseline.
- **State Structure**: $\mathcal{O}(1)$ sliding double-ended queue (`collections.deque`) maintaining (`event_time_epoch`, `amount`) per user.

4.2 Derived Features & Velocity Ratios

For every incoming transaction, Flink computes:

1. `tx_count_5m`: Number of transactions in the last 5 minutes.
2. `tx_sum_5m`: Total expenditure in the last 5 minutes.
3. `tx_count_10m`: Number of transactions in the last 10 minutes.
4. `tx_sum_10m`: Total expenditure in the last 10 minutes.
5. `tx_velocity_ratio`: $\frac{\text{tx_sum_5m}}{\max(\text{tx_sum_10m}, 1.0)}$ (burst ratio indicator).
6. `cep_alert`: Boolean trigger when `tx_count_5m >= 3` and `tx_velocity_ratio >= 0.7`.

4.3 Output Schema (`fraud-features`)

```
{
  "transaction_id": "TX12345",
  "user_id": "USER123",
  "card_id": "CARD123",
  "amount": 1250.50,
  "event_time": "2026-10-05T11:55:00Z",
  "tx_count_5m": 4,
```

```

"tx_sum_5m": 3450.00,
"tx_count_10m": 5,
"tx_sum_10m": 3750.00,
"tx_velocity_ratio": 0.92,
"cep_alert": true,
"processed_at": "2026-10-05T11:55:00.125Z"
}

```

5. Apache Spark Batch & Streaming Engine (Member 2)

5.1 Spark Batch Processing Engine

- **Target:** Complete 590,540-record IEEE-CIS dataset.
- **Vectorized Parquet Engine:** Pre-computed column-projected Snappy Parquet representation.
- **Aggregation Pipeline:** Multi-column fraud distributions, card-level velocity statistics, amount quantiles, and missing value profiling.
- **Performance:**
 - Unoptimized CSV Baseline: **25.71s** (22,970 records/sec)
 - Vectorized Optimized Parquet: **0.822s (718,573 records/sec $\rightarrow$ 31.28x speedup)**

5.2 Spark Streaming Consumer Engine

- **Consumer Group:** `spark-streaming-fraud-group`
- **Topic:** `fraud-transactions`
- **Micro-Batch Inference:** Uses pre-trained LightGBM model pipeline (`spark/models/`) to compute instant fraud risk probabilities ($P(\text{fraud})$).
- **Latency:** Sub-millisecond inference per micro-batch.

6. End-to-End Integration Architecture

The 14-step integration test framework validates every stage of the pipeline:

```

[Step 1]  Kafka Broker Health Check (localhost:9092)
[Step 2]  Topic Topology & Partition Verification
[Step 3]  Canonical Schema Validator Initialization
[Step 4]  Synthetic & IEEE-CIS Transaction Ingestion
[Step 5]  Kafka Producer acks=all Execution
[Step 6]  Message Partition Routing Audit (card_id hash)
[Step 7]  Flink Consumer Initialization
[Step 8]  Flink Stateful keyBy(user_id) Processing
[Step 9]  5m & 10m Sliding Window Feature Calculation
[Step 10] Flink Sink Delivery to fraud-features Topic
[Step 11] Spark Streaming Consumer Ingestion (spark-streaming-fraud-group)
[Step 12] Spark Real-Time Feature Scoring & Inference
[Step 13] Spark Batch Historical Processing (590,540 Records)
[Step 14] End-to-End Zero-Loss Count & Integrity Reconciliation

```

7. Comprehensive Performance & Benchmark Results

7.1 Verified Execution Metrics Summary

Component	Dataset / Scale	Records Processed	Execution Time	Throughput	Zero-Loss Status
Spark Batch (Optimized)	Historical IEEE-CIS	590,540 rows	0.822 s	718,573.9 rec/s	PASS (Exact 20,663 fraud)
Spark Batch (Baseline)	Historical IEEE-CIS CSV	590,540 rows	25.71 s	22,970.0 rec/s	PASS
Flink CEP (1,000)	Kafka Stream	1,000 events	0.134 s	7,462.7 rec/s	PASS (1,000/1,000 delivered)
Flink CEP (5,000)	Kafka Stream	5,000 events	0.551 s	9,074.4 rec/s	PASS (5,000/5,000 delivered)
Flink CEP (10,000)	Kafka Stream	10,000 events	1.048 s	9,540.2 rec/s	PASS (10,000/10,000 delivered)
Spark Streaming (1,000)	Kafka Micro-batch	1,000 events	0.063 s	15,873.0 rec/s	PASS (1,000/1,000 consumed)
Spark Streaming (5,000)	Kafka Micro-batch	5,000 events	0.256 s	19,531.2 rec/s	PASS (5,000/5,000 consumed)
Spark Streaming (10,000)	Kafka Micro-batch	10,000 events	0.493 s	20,300.6 rec/s	PASS (10,000/10,000 consumed)
12P $\times$ 12C Scaled Pipeline	Full IEEE-CIS Stream	590,540 events	Concurrent	12-Consumer Balanced	PASS (Uniform distribution)

8. Complete Repository Directory & File Inventory

Apache-Kafka/	
├─ COMPLETE_CODEBASE_GUIDE.md	# Master comprehensive system guide
(this file)	
├─ CODEBASE_INPUT_OUTPUT.md	# Detailed I/O format specifications
├─ KAFKA_SPARK_FLINK_INTEGRATION_SPEC.md	# Inter-team contract and technical
interface specs	
├─ END_TO_END_INTEGRATION_REPORT.md	# 14-Step formal integration test
report	
├─ OPTIMIZATION_AND_BENCHMARK_REPORT.md	# Multi-scale optimization &
benchmark report	

```

└─ run_full_e2e_integration_test.py      # Automated 14-step integration test
runner
└─ test_scale_benchmark_optimizations.py # Multi-scale benchmark suite (1k,
5k, 10k)
└─ run_kafka_spark_flink_pipeline.py     # Live multi-process end-to-end
streaming pipeline
└─ run_full_ieee_cis_12p_12c_benchmark.py # 12-Partition x 12-Consumer full
590k benchmark
└─ run_spark_batch_benchmark.py          # Standalone Spark Batch benchmark
└─ run_spark_streaming_benchmark.py      # Standalone Spark Streaming
benchmark
└─ run_flink_benchmark.py               # Standalone Flink CEP benchmark
└─ verify_spark_batch_ieee_cis.py        # Historical IEEE-CIS data
verification script
|
└─ kafka/                               # Member 1: Kafka Ingestion &
Producer Layer
|   └─ config/
|       └─ kafka_config.py              # Central Kafka broker, topic, and
consumer settings
|           └─ logging_config.py        # Structured logging configuration
|               └─ producer/
|                   └─ transaction_producer.py # Core Kafka Producer with acks=all &
card_id keying
|                       └─ generator.py   # Synthetic transaction event
generator
|   └─ multi_dataset_producer.py         # Ingestion wrapper for IEEE-CIS, CC-
10k, PaySim
|       └─ consumer/
|           └─ flink_consumer.py        # Flink CEP Consumer with
keyBy(user_id) & sliding windows
|               └─ spark_streaming_consumer.py # Spark Streaming Consumer (spark-
streaming-fraud-group)
|                   └─ transaction_consumer.py # Base Kafka consumer utility
|                       └─ tests/
|                           └─ test_spark_flink_integration.py # Member 1 <-> Member 2 contract unit
tests
|   └─ test_producer.py                 # Producer unit tests
|       └─ test_consumer.py            # Consumer unit tests
|           └─ utils/
|               └─ dataset_loader.py    # CSV / Parquet data loaders
|                   └─ metrics.py       # Latency, throughput & lag
measurement tools
|   └─ helpers.py                      # Serialization and formatting
helpers
|
└─ spark/                               # Member 2: Spark Batch, Streaming &
Flink Models
|   └─ batch/
|       └─ batch_processor.py          # Spark Batch analytics engine
|           └─ optimize_batch.py       # Vectorized Parquet optimization
routines
|   └─ streaming/
|       └─ stream_processor.py         # Structured Streaming feature

```

```

extraction
├── feature_pipeline.py           # Real-time feature calculation
├── models/
│   ├── fraud_detection_model.py # LightGBM fraud classification model
│   └── model_evaluator.py       # Evaluation metrics (ROC-AUC,
Precision-Recall)
└── spark-benchmark-results/     # Historical benchmark archives &
logs

```

9. Step-by-Step Execution & Reproduction Guide

All commands are validated for Windows PowerShell in the project workspace directory:

9.1 Environment Setup

```

# Activate Python Virtual Environment
& "C:/Users/HADASSAH KIRAN/OneDrive/Documents/GitHub/Reaserch-Work/Apache-
Kafka/.venv/Scripts/Activate.ps1"

```

9.2 Running the Full 14-Step End-to-End Integration Suite

```
python run_full_e2e_integration_test.py
```

Validates broker health, partition routing, Flink CEP sliding windows, Spark Streaming micro-batches, Spark Batch historical processing, and zero data loss.

9.3 Running the Multi-Scale Benchmark Suite (1,000, 5,000, and 10,000 Events)

```
python test_scale_benchmark_optimizations.py
```

Executes isolated benchmarks across Spark Batch, Flink CEP, and Spark Streaming with automated throughput calculation.

9.4 Running the 12-Partition $\times$ 12-Consumer Full IEEE-CIS Scale Benchmark

```
python run_full_ieee_cis_12p_12c_benchmark.py
```

*Creates topic ***fraud-transactions-p12***, spins up 12 parallel consumer workers, and streams the entire 590,540 IEEE-CIS dataset.*

9.5 Running the Live Multi-Process Streaming Pipeline

```
python run_kafka_spark_flink_pipeline.py
```

Launches Kafka Producer, Flink CEP Engine, and Spark Streaming Consumer concurrently with real-time log aggregation.

Documentation generated for research repository: [PeramkondaIshwarya/Apache-Kafka](#).